

Chungpa Lee

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EDUCATION

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| Mar 2021 – Present | Yonsei University , Seoul, Republic of Korea
Ph.D. Candidate in Statistics and Data Science
GPA: 4.22/4.30 (4.00/4.00)
<i>Expected to graduate in Aug 2027</i> |
| Mar 2015 – Feb 2021 | Yonsei University , Seoul, Republic of Korea
Bachelor of Arts in Applied Statistics, Minor in Economics
Overall GPA: 3.93/4.30 (3.77/4.00), Major GPA in Applied Statistics: 4.13/4.30 (3.92/4.00)
<i>Leave of Absence for Military Service (Feb 2017 – Nov 2018)</i> |

WORK EXPERIENCE

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| Aug 2026 – Present | VISIT Intern (<i>visiting student</i>), Paul G. Allen School of Computer Science and Engineering, University of Washington
Host: Prof. Sewoong Oh
– Conduct research on language models |
| May 2025 – Apr 2026 | Honorary Associate (<i>visiting researcher</i>), Department of Electrical and Computer Engineering, University of Wisconsin–Madison
Host: Prof. Kangwook Lee
– Conducted research on language models |
| May 2022 – Sep 2024 | Manager , Data Science Team, GRETA Inc.
Directors: Prof. Joseph H.T. Kim and Prof. Jongho Im
– Led the development of a synthetic data generation solution (Mar 2023 – Sep 2024)
– Developed a forecasting model for auto parts at Hyundai Mobis (Sep 2022 – Feb 2023)
– Developed a synthetic data generation solution for CRO companies (Jun 2022 – Feb 2023)
– Developed a statistical analysis solution (part-time; Jul 2021 – Dec 2021) |
| Mar 2023 – Aug 2024 | Statistical Consulting Assistant , Yonsei Institute of Data Science
Director: Prof. Seungho Kang
– Provided statistical consulting for academic research and completed over 20 projects |

RESEARCH INTERESTS

My research develops theoretical frameworks to explain when and why machine learning succeeds or fails and to guide principled methods. To this end, I identify the limitations of existing approaches and develop theoretically grounded solutions. My current research focuses on **language models**, **representation learning**, and **synthetic data generation**.

PUBLICATIONS

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- [1] *How to Correctly Report LLM-as-a-Judge Evaluations*
Chungpa Lee, Thomas Zeng, Jongwon Jeong, Jy-yong Sohn, Kangwook Lee
 Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026
 - [2] *Fine-Tuning Without Forgetting In-Context Learning: A Theoretical Analysis of Linear Attention Models*
Chungpa Lee, Jy-yong Sohn, Kangwook Lee
 Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026

- [3] *Transformers in the Dark: Navigating Unknown Search Spaces via Bandit Feedback*
Jungtaek Kim, Thomas Zeng, Ziqian Lin, Minjae Lee, **Chungpa Lee**, Jy-yong Sohn, Hyung Il Koo, Kangwook Lee
Transactions on Machine Learning Research (TMLR), 2026
- [4] *On the Similarities of Embeddings in Contrastive Learning*
Chungpa Lee, Sehee Lim, Kibok Lee, Jy-yong Sohn
Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025
- [5] *A Generalized Theory of Mixup for Structure-Preserving Synthetic Data*
Chungpa Lee, Jongho Im, Joseph H.T. Kim
Proceedings of The 28th International Conference on Artificial Intelligence and Statistics (AISTATS), 2025
- [6] *A Theoretical Framework for Preventing Class Collapse in Supervised Contrastive Learning*
Chungpa Lee, Jeongheon Oh, Kibok Lee, Jy-yong Sohn
Proceedings of The 28th International Conference on Artificial Intelligence and Statistics (AISTATS), 2025
- [7] *Analysis of Using Sigmoid Loss for Contrastive Learning*
Chungpa Lee, Joonhwan Chang, Jy-yong Sohn
Proceedings of The 27th International Conference on Artificial Intelligence and Statistics (AISTATS), 2024

WORKING PAPERS

- [1] *Same Concept, Different Directions: Cross-Modal Feature Heterogeneity in Sparse Autoencoders*
Chungpa Lee^{*}, Jihoon Kwon^{*}, Kyle Min, Jy-yong Sohn (**equal contribution*)
(under review)
- [2] *Rethinking Representation Geometry in Supervised Contrastive Learning under Long-Tailed Data*
JoonHyeok Jeong, **Chungpa Lee**, Jy-yong Sohn, Kibok Lee
(under review)
- [3] *From Local Mixing to Global Consistency via Barycentric Calibration*
Nathaniel Kang, **Chungpa Lee**
(under review)
- [4] *Poisson Regression with Additive Exponential Mean: Statistical Modeling and Insurance Applications*
Chungpa Lee, Joseph H.T. Kim
(under review)

HONORS AND AWARDS

Aug 2026 – Aug 2027	Shin Dongwook Study Abroad (<i>scholarship</i>), Yonsei University
May 2025 – Apr 2026	BK21 FOUR International Joint Training Support for Outstanding Graduate Students, National Research Foundation of Korea
Spring 2025, Fall 2021	Academy Research Fellowship, Yonsei University
Mar 2021 – Feb 2023	Yongwoon Scholarship, Yongwoon Scholarship Foundation
Fall 2020, Spring 2015	Semester Honors (<i>undergraduate</i>), Yonsei University
Spring 2020	Semester Highest Honors (<i>undergraduate</i>), Yonsei University

TEACHING EXPERIENCE

- Mar 2021 – Feb 2025 **Teaching Assistant, Yonsei University**
Department of Statistics and Data Science
– STA9200 Advanced Deep Learning Theory (Fall 2024)
– STA9403 Big Data Analysis For Industrial Application (Spring 2022, Fall 2022)
Department of Applied Statistics
– STA3125 Regression Analysis (Spring 2021, Fall 2021, Spring 2022, Fall 2022)
– STA1001 Introduction to Statistics (Spring 2021, Fall 2021, Spring 2022)
– STA4111 Statistics for Insurance (Spring 2022)
Graduate School of Economics
– RC597 Understanding the Global Economy (Fall 2021)

ACADEMIC SERVICE

Reviewer:

- **Conferences:** ICML 2026, NeurIPS 2026, AAAI 2027
- **Journals:** IEEE Transactions on Neural Networks and Learning Systems
- **Magazines:** IEEE BITS the Information Theory Magazine
- **Workshops:** Mechanistic Interpretability Workshop at ICML 2026, Trust-AI-Eval Workshop at NeurIPS 2026

LANGUAGES AND SKILLS

Languages: English (*fluent*), Korean (*native*)

Technical Skills: Python, PyTorch, R, Repp, Julia, SQL, Git, Docker, Bash

REFERENCES

1. **Joseph H.T. Kim**
Professor, Department of Statistics and Data Science, Yonsei University
✉ jhtkim@yonsei.ac.kr
2. **Jy-yong Sohn**
Associate Professor, Department of Statistics and Data Science, Yonsei University
✉ jysohn1108@yonsei.ac.kr 🌐 <https://itml.yonsei.ac.kr/>
3. **Kangwook Lee**
Chief AI Officer, KRAFTON
Chief Technology Officer, Ludo Robotics
✉ kangwook.lee@krafton.com 🌐 <https://kangwooklee.com/>